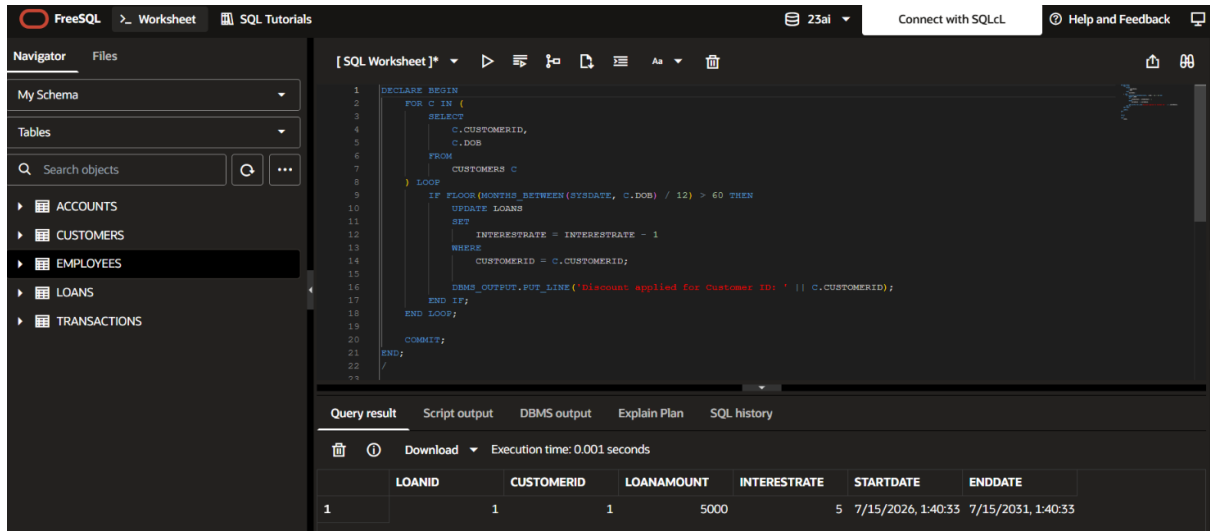


PLSQL_Exercises

Exercise 1: Control Structures

Scenario 1 :



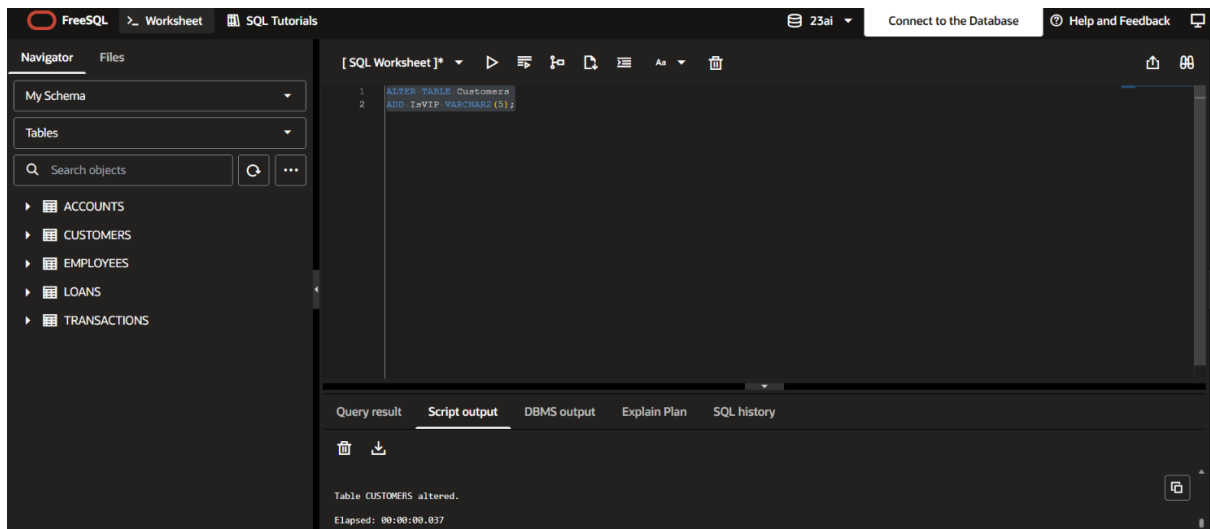
The screenshot shows the FreeSQL IDE interface. The left sidebar contains a Navigator pane with a tree view of database objects: ACCOUNTS, CUSTOMERS, EMPLOYEES, LOANS, and TRANSACTIONS. The main editor displays a PL/SQL script with the following code:

```
1 DECLARE BEGIN
2   FOR C IN (
3     SELECT
4       C.CUSTOMERID,
5       C.DOB
6     FROM
7       CUSTOMERS C
8   ) LOOP
9     IF FLOOR(MONTHS_BETWEEN(SYSDATE, C.DOB) / 12) > 60 THEN
10      UPDATE LOANS
11      SET
12        INTERESTRATE = INTERESTRATE - 1
13      WHERE
14        CUSTOMERID = C.CUSTOMERID;
15      DBMS_OUTPUT.PUT_LINE('Discount applied for Customer ID: ' || C.CUSTOMERID);
16    END IF;
17  END LOOP;
18 COMMIT;
19 END;
```

Below the editor, the 'Query result' tab is active, showing the execution output. The execution time is 0.001 seconds. The result is a table with the following data:

LOANID	CUSTOMERID	LOANAMOUNT	INTERESTRATE	STARTDATE	ENDDATE
1	1	1	5000	5 7/15/2026, 1:40:33	7/15/2031, 1:40:33

Scenario 2 :



The screenshot shows the FreeSQL IDE interface. The left sidebar contains a Navigator pane with a tree view of database objects: ACCOUNTS, CUSTOMERS, EMPLOYEES, LOANS, and TRANSACTIONS. The main editor displays an SQL script with the following code:

```
1 ALTER TABLE Customers
2 ADD IsVIP VARCHAR(5);
```

Below the editor, the 'Script output' tab is active, showing the execution output. The output is:

```
Table CUSTOMERS altered.
Elapsed: 00:00:00.037
```

Navigator Files

My Schema

Tables

Search objects

ACCOUNTS

CUSTOMERS

EMPLOYEES

LOANS

TRANSACTIONS

[SQL Worksheet]*

```
3
4 BEGIN
5
6 FOR c IN (SELECT CustomerID, Balance FROM Customers)
7
8 LOOP
9
10 IF c.Balance > 10000 THEN
11
12 UPDATE Customers
13
14 SET IsVIP='TRUE'
15
16 WHERE CustomerID=c.CustomerID;
17
18 DBMS_OUTPUT.PUT_LINE('VIP Customer: '||c.CustomerID);
19
20 END IF;
21
22 END LOOP;
23
24 COMMIT;
25
26 END;
27
28 /
```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.006 seconds

	CUSTOMERID	NAME	BALANCE	ISVIP
1	1	John Doe	1000	(null)
2	2	Jane Smith	1500	(null)

Scenario 3 : If no loan is due within 30 days, it prints nothing.

FreeSQL > Worksheet SQL Tutorials 23ai Connect to the Database

Navigator Files

My Schema

Tables

Search objects

ACCOUNTS

CUSTOMERS

EMPLOYEES

LOANS

TRANSACTIONS

[SQL Worksheet]*

```
1 BEGIN
2
3 FOR l IN (
4
5 SELECT LoanID,
6 CustomerID,
7 EndDate
8
9 FROM Loans
10
11 WHERE EndDate BETWEEN SYSDATE
12 AND SYSDATE-30
13
14 )
15
16 LOOP
17
18 DBMS_OUTPUT.PUT_LINE(
19 'Reminder: Customer '
20 ||l.CustomerID||
21 ' loan '
22 ||l.LoanID||
23 ' is due soon.'
24 );
25
26 END LOOP;
27
28 END;
29 /
```

Query result Script output DBMS output Explain Plan SQL history

Show more...

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.002

Exercise 3: Stored Procedures

Scenario 1 :

NavigatorFiles

My Schema

Tables

Search objects

ACCOUNTS

CUSTOMERS

EMPLOYEES

LOANS

TRANSACTIONS

[SQL Worksheet]*

1CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest

2

3IS

4

5BEGIN

6

7UPDATE Accounts

8

9SET Balance=Balance+(Balance*0.01)

10

11WHERE AccountType='Savings';

12

13COMMIT;

14

15END;

16/

Query resultScript outputDBMS outputExplain PlanSQL history

Procedure PROCESSMONTHLYINTEREST compiled

Elapsed: 00:00:00.017

NavigatorFiles

My Schema

Tables

Search objects

ACCOUNTS

CUSTOMERS

EMPLOYEES

LOANS

TRANSACTIONS

[SQL Worksheet]*

1CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest

2

3IS

4

5BEGIN

6

7UPDATE Accounts

8

9SET Balance=Balance+(Balance*0.01)

10

11WHERE AccountType='Savings';

12

13COMMIT;

14

15END;

16/

17SELECT * FROM Accounts;

Query resultScript outputDBMS outputExplain PlanSQL history

DownloadExecution time: 0.009 seconds

	ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE	LASTMODIFIED
1		1	1 Savings	1000	7/15/2026, 1:40:33
2		2	2 Checking	1500	7/15/2026, 1:40:33

Scenario 2 :

Navigator **Files**

My Schema

Tables

Search objects

- ACCOUNTS
- CUSTOMERS
- EMPLOYEES
- LOANS
- TRANSACTIONS

[SQL Worksheet]*

```
1 CREATE OR REPLACE PROCEDURE UPDATEEMPLOYEEBONUS (
2   P_DEPARTMENT VARCHAR2,
3   P_BONUS       NUMBER
4 ) IS
5 BEGIN
6   UPDATE EMPLOYEES
7   SET
8     SALARY = SALARY + ( SALARY * P_BONUS / 100 )
9   WHERE
10    DEPARTMENT = P_DEPARTMENT;
11  COMMIT;
12 END;
13 /
14
15 BEGIN
16   UPDATEEMPLOYEEBONUS('HR', 10);
17 END;
18 /
19
20 SELECT
21   EMPLOYEEID,
22   NAME,
23   SALARY
24 FROM
25   EMPLOYEES;
```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.004 seconds

	EMPLOYEEID	NAME	SALARY
1	1	Alice Johnson	84700
2	2	Bob Brown	60000

Scenario 3:

Navigator **Files**

My Schema

Tables

Search objects

- ACCOUNTS
- CUSTOMERS
- EMPLOYEES
- LOANS
- TRANSACTIONS

[SQL Worksheet]*

```
1 CREATE OR REPLACE PROCEDURE TransferFunds(
2   p_from NUMBER,
3   p_to NUMBER,
4   p_amount NUMBER
5 )
6 IS
7   v_from_balance NUMBER;
8   v_to_balance NUMBER;
9 BEGIN
10  SELECT Balance INTO v_from_balance
11  FROM Accounts
12  WHERE AccountID = p_from;
13
14  SELECT Balance INTO v_to_balance
15  FROM Accounts
16  WHERE AccountID = p_to;
17
18  DBMS_OUTPUT.PUT_LINE('Before Transfer:');
19  DBMS_OUTPUT.PUT_LINE('From Account Balance = ' || v_from_balance);
20  DBMS_OUTPUT.PUT_LINE('To Account Balance = ' || v_to_balance);
21
22  IF v_from_balance >= p_amount THEN
23    UPDATE Accounts
24    SET Balance = Balance - p_amount
25    WHERE AccountID = p_from;
26
27    UPDATE Accounts
28    SET Balance = Balance + p_amount
29    WHERE AccountID = p_to;
30
31    COMMIT;
32
33    SELECT Balance INTO v_from_balance
34    FROM Accounts
35    WHERE AccountID = p_from;
36
37    SELECT Balance INTO v_to_balance
38    FROM Accounts
39    WHERE AccountID = p_to;
40
41    DBMS_OUTPUT.PUT_LINE('After Transfer:');
42    DBMS_OUTPUT.PUT_LINE('From Account Balance = ' || v_from_balance);
43    DBMS_OUTPUT.PUT_LINE('To Account Balance = ' || v_to_balance);
44  ELSE
45    DBMS_OUTPUT.PUT_LINE('Insufficient Balance');
46  END IF;
47 END;
48 /
```

Query resultScript outputDBMS outputExplain PlanSQL history

p_to NUMBER,
p_amount NUMBER...

Show more...

Procedure TRANSFERFUNDS compiled

Elapsed: 00:00:00.023

NavigatorFiles

My Schema

Tables

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▶

ACCOUNTS

▶

CUSTOMERS

▶

EMPLOYEES

▶

LOANS

▶

TRANSACTIONS

[SQL Worksheet]*

Aa

1

BEGIN

2

TransferFunds (1, 2, 200);

3

END;

4

/

Query resultScript outputDBMS outputExplain Plan

Before Transfer:
From Account Balance = 600
To Account Balance = 1900
After Transfer:
From Account Balance = 400
To Account Balance = 2100

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.012